

```
In [5]: import numpy as np
```

```
In [6]: np.__version__
```

```
Out[6]: '2.1.3'
```

```
In [7]: np.__version__
```

```
Out[7]: '2.1.3'
```

```
In [8]: import sys  
sys.version
```

```
Out[8]: '3.13.5 | packaged by Anaconda, Inc. | (main, Jun 12 2025, 16:37:03) [MSC v.192  
9 64 bit (AMD64)]'
```

```
In [9]: my_list=[0,1,2,3,4,5]  
my_list
```

```
Out[9]: [0, 1, 2, 3, 4, 5]
```

```
In [10]: arr = np.array(my_list)
```

```
In [11]: arr
```

```
Out[11]: array([0, 1, 2, 3, 4, 5])
```

```
In [12]: type(arr)
```

```
Out[12]: numpy.ndarray
```

```
In [13]: type(my_list)
```

```
Out[13]: list
```

```
In [14]: np.arange(10)
```

```
Out[14]: array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [15]: list(range(10))
```

```
Out[15]: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

```
In [16]: r=([6,5,4,3])
```

```
print(type(r))
```

```
<class 'list'>
```

```
In [17]: rr=np.array2string(arr)  
print(rr)  
print(type(rr))
```

```
[0 1 2 3 4 5]
```

```
<class 'str'>
```

```
print(type(rr))
```

```
In [18]: np.arange(20)
```

```
Out[18]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
                17, 18, 19])
```

```
In [20]: np.arange(5.0)
```

```
Out[20]: array([0., 1., 2., 3., 4.])
```

```
In [21]: np.arange(0,4)
```

```
Out[21]: array([0, 1, 2, 3])
```

```
In [22]: np.arange(10,20)
```

```
Out[22]: array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
```

```
In [23]: np.arange(20,10,) # 1st arg < 2nd arg
```

```
Out[23]: array([], dtype=int64)
```

```
In [24]: np.arange(-20,10)
```

```
Out[24]: array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10, -9, -8,
                -7, -6, -5, -4, -3, -2, -1,  0,  1,  2,  3,  4,  5,
                 6,  7,  8,  9])
```

```
In [26]: np.arange(10,50,5) # allow only 3 arg
```

```
Out[26]: array([10, 15, 20, 25, 30, 35, 40, 45])
```

```
In [27]: np.zeros(3) # parameter 0.
```

```
Out[27]: array([0., 0., 0.])
```

```
In [28]: np.zeros(3,dtype=int) # hyparparameter 0.
```

```
Out[28]: array([0, 0, 0])
```

```
In [30]: np.zeros([2,2],dtype=int)
```

```
Out[30]: array([[0, 0],
                [0, 0]])
```

```
In [31]: np.ones(3)
```

```
Out[31]: array([1., 1., 1.])
```

```
In [32]: np.ones([10,3],dtype=int)
```

```
Out[32]: array([[1, 1, 1],
                [1, 1, 1],
                [1, 1, 1],
                [1, 1, 1],
                [1, 1, 1],
                [1, 1, 1],
                [1, 1, 1],
                [1, 1, 1],
                [1, 1, 1],
                [1, 1, 1]])
```

```
In [33]: np.two(3)
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[33], line 1
----> 1 np.twos(3)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:414, in __getattr__(attr)
    411     import numpy.char as char
    412     return char.chararray
--> 414 raise AttributeError("module {!r} has no attribute "
    415                        "{!r}".format(__name__, attr))

AttributeError: module 'numpy' has no attribute 'twos'
```

```
In [34]: rand(3,2)
```

```
-----
NameError                                    Traceback (most recent call last)
Cell In[34], line 1
----> 1 rand(3,2)

NameError: name 'rand' is not defined
```

```
In [35]: random.rand(3,2)
```

```
-----
NameError                                    Traceback (most recent call last)
Cell In[35], line 1
----> 1 random.rand(3,2)

NameError: name 'random' is not defined
```

```
In [37]: np.random.rand(3,2)
```

```
Out[37]: array([[0.72892456, 0.35160127],
                [0.39035542, 0.88110825],
                [0.07946144, 0.20336887]])
```

```
In [39]: np.random.randint(2)
```

```
Out[39]: 1
```

```
In [43]: np.random.randint(4)
```

```
Out[43]: 3
```

```
In [45]: np.random.randint(1,8,100) #creates 50 num betw 1 to 7
```

```
Out[45]: array([7, 5, 2, 5, 7, 4, 7, 1, 3, 4, 4, 7, 2, 6, 4, 7, 6, 7, 2, 1, 7, 4,
        6, 1, 6, 3, 3, 7, 2, 4, 2, 3, 7, 3, 1, 7, 5, 6, 6, 5, 2, 4, 7, 7,
        2, 1, 3, 5, 4, 4, 7, 5, 3, 5, 5, 5, 3, 7, 3, 2, 4, 2, 3, 4, 7, 6,
        3, 3, 7, 5, 5, 5, 3, 2, 2, 3, 3, 6, 2, 2, 5, 7, 4, 1, 5, 2, 3, 1,
        1, 3, 7, 4, 5, 3, 1, 7, 1, 1, 2, 5], dtype=int32)
```

```
In [46]: np.random.randint(10,40,(10,10))
```

```
Out[46]: array([[26, 14, 18, 18, 22, 16, 39, 19, 21, 38],
        [18, 32, 28, 30, 14, 14, 32, 39, 28, 39],
        [21, 31, 18, 19, 33, 15, 32, 13, 27, 38],
        [26, 34, 23, 35, 16, 24, 21, 22, 25, 25],
        [14, 33, 18, 32, 15, 32, 11, 35, 33, 31],
        [34, 23, 25, 35, 31, 18, 39, 37, 22, 36],
        [10, 13, 21, 35, 20, 27, 19, 34, 18, 17],
        [10, 18, 39, 16, 27, 37, 10, 20, 31, 27],
        [20, 28, 23, 28, 10, 15, 11, 19, 29, 37],
        [32, 25, 12, 21, 30, 17, 26, 10, 38, 12]], dtype=int32)
```

```
In [47]: np.arange(1,13).reshape(3,4)
```

```
Out[47]: array([[ 1,  2,  3,  4],
        [ 5,  6,  7,  8],
        [ 9, 10, 11, 12]])
```

```
In [48]: np.arange(1,13).reshape(3,5)
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[48], line 1
----> 1 np.arange(1,13).reshape(3,5)

ValueError: cannot reshape array of size 12 into shape (3,5)
```

slicing in matrix

```
In [49]: b=np.random.randint(10,20,(5,4))
        b
```

```
Out[49]: array([[10, 14, 19, 11],
        [12, 16, 18, 14],
        [11, 16, 18, 18],
        [17, 18, 14, 15],
        [16, 18, 18, 16]], dtype=int32)
```

```
In [50]: b[:]
```

```
Out[50]: array([[10, 14, 19, 11],
        [12, 16, 18, 14],
        [11, 16, 18, 18],
        [17, 18, 14, 15],
        [16, 18, 18, 16]], dtype=int32)
```

```
In [51]: b[0]
```

```
Out[51]: array([10, 14, 19, 11], dtype=int32)
```

```
In [52]: b[0:5]
```

```
Out[52]: array([[10, 14, 19, 11],
               [12, 16, 18, 14],
               [11, 16, 18, 18],
               [17, 18, 14, 15],
               [16, 18, 18, 16]], dtype=int32)
```

```
In [54]: b[0,1]
```

```
Out[54]: np.int32(14)
```

```
In [55]: b[::-1]
```

```
Out[55]: array([[16, 18, 18, 16],
               [17, 18, 14, 15],
               [11, 16, 18, 18],
               [12, 16, 18, 14],
               [10, 14, 19, 11]], dtype=int32)
```

```
In [57]: print(b[::-2])
```

```
[[16 18 18 16]
 [11 16 18 18]
 [10 14 19 11]]
```

```
In [58]: b[-3,-3]
```

```
Out[58]: np.int32(16)
```

```
In [59]: b.max()
```

```
Out[59]: np.int32(19)
```

```
In [60]: b.min()
```

```
Out[60]: np.int32(10)
```

```
In [61]: b.mean()
```

```
Out[61]: np.float64(15.45)
```

```
In [62]: b
```

```
Out[62]: array([[10, 14, 19, 11],
               [12, 16, 18, 14],
               [11, 16, 18, 18],
               [17, 18, 14, 15],
               [16, 18, 18, 16]], dtype=int32)
```

```
In [63]: b[1]
```

```
Out[63]: array([12, 16, 18, 14], dtype=int32)
```

```
In [64]: b[2:3,0:2]
```

```
Out[64]: array([[11, 16]], dtype=int32)
```

In [65]: `b[1:4,1:4]`

Out[65]: `array([[16, 18, 14],
[16, 18, 18],
[18, 14, 15]], dtype=int32)`

In [66]: `mat`

```
-----
NameError                                Traceback (most recent call last)
Cell In[66], line 1
----> 1 mat

NameError: name 'mat' is not defined
```

In [67]: `mat=np.random.randint(1,100,(10,10))`

In [68]: `mat`

Out[68]: `array([[55, 37, 38, 77, 77, 39, 1, 74, 79, 60],
[38, 24, 83, 50, 90, 63, 25, 8, 9, 34],
[89, 36, 81, 60, 57, 19, 44, 88, 31, 86],
[16, 75, 11, 91, 17, 85, 21, 70, 65, 87],
[22, 1, 21, 38, 3, 28, 86, 96, 26, 29],
[72, 23, 62, 95, 88, 86, 97, 78, 80, 76],
[11, 42, 5, 99, 10, 79, 48, 60, 81, 40],
[48, 96, 33, 63, 8, 32, 44, 41, 55, 23],
[20, 36, 3, 57, 43, 90, 75, 90, 25, 16],
[57, 7, 57, 83, 73, 34, 62, 73, 3, 86]], dtype=int32)`

In [69]: `mat>50`

Out[69]: `array([[True, False, False, True, True, False, False, True, True,
 True],
[False, False, True, False, True, True, False, False, False,
 False],
[True, False, True, True, True, False, False, True, False,
 True],
[False, True, False, True, False, True, False, True, True,
 True],
[False, False, False, False, False, False, True, True, False,
 False],
[True, False, True, True, True, True, True, True, True,
 True],
[False, False, False, True, False, True, False, True, True,
 False],
[False, True, False, True, False, False, False, False, True,
 False],
[False, False, False, True, False, True, True, True, False,
 False],
[True, False, True, True, True, False, True, True, False,
 True]])`

In [82]: `mask=(mat>50)
mask_int = mask.astype(int)
result=mat[mask]
print(mask_int)
print(mask)`

```
[[1 0 0 1 1 0 0 1 1 1]
 [0 0 1 0 1 1 0 0 0 0]
 [1 0 1 1 1 0 0 1 0 1]
 [0 1 0 1 0 1 0 1 1 1]
 [0 0 0 0 0 0 1 1 0 0]
 [1 0 1 1 1 1 1 1 1 1]
 [0 0 0 1 0 1 0 1 1 0]
 [0 1 0 1 0 0 0 0 1 0]
 [0 0 0 1 0 1 1 1 0 0]
 [1 0 1 1 1 0 1 1 0 1]]
[[ True False False  True  True False False  True  True  True]
 [False False  True False  True  True False False False False]
 [ True False  True  True  True False False  True False  True]
 [False  True False  True False  True False  True  True  True]
 [False False False False False False  True  True False False]
 [ True False  True  True  True  True  True  True  True  True]
 [False False False  True False  True False  True  True False]
 [False  True False  True False False False False  True False]
 [False False False  True False  True  True  True False False]
 [ True False  True  True  True False  True  True False  True]]
```

In []: